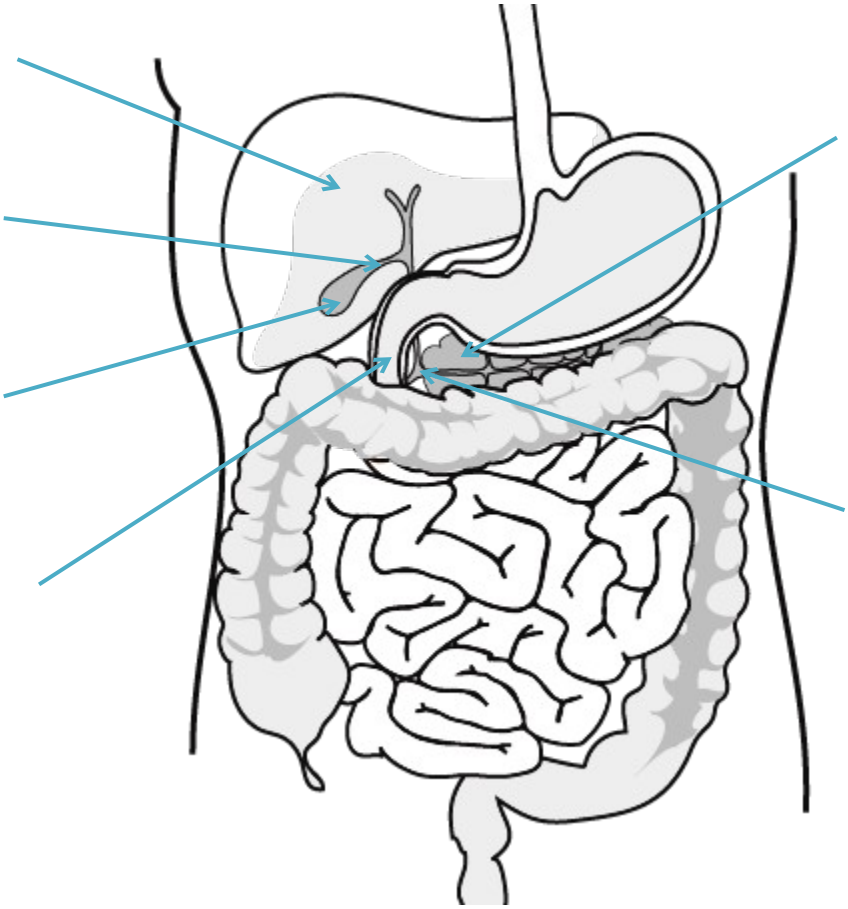
Task 1

Colour in the diagram using the following key:

liver	**red**	gall bladder	**green**
pancreas	**yellow**	duodenum	**blue**

Label the diagram using the table below.



liver produces bile	bile duct	pancreatic duct
duodenum	bile stored in gall bladder	digestive enzymes produced in pancreas

Task 2

1. What role does bile have in digestion?

- It neutralises food, coming from the ....., which is ..... .

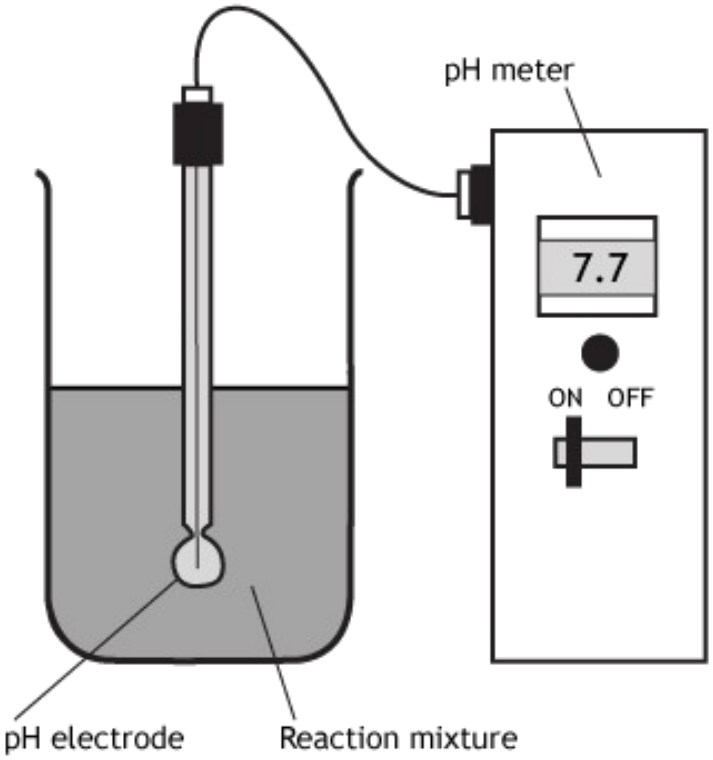
- It provides an ..... environment – which is the ..... pH for the enzymes which work in the small intestine.

- It emulsifies ..... – increasing the surface area of the fat exposed to ..... enzymes.

2. Draw a diagram to illustrate the role of bile in digestion. Don't add labels, but swap your diagram with your partner and label their diagram.



Task 3 – digesting milk fat



The diagram shows the apparatus used to investigate the digestion of milk fat by an enzyme. The reaction mixture contained milk and the enzyme.

- In **experiment 1**, bile was added as well.
- In **experiment 2**, water was added instead of bile.

The pH was recorded every 2 minutes.

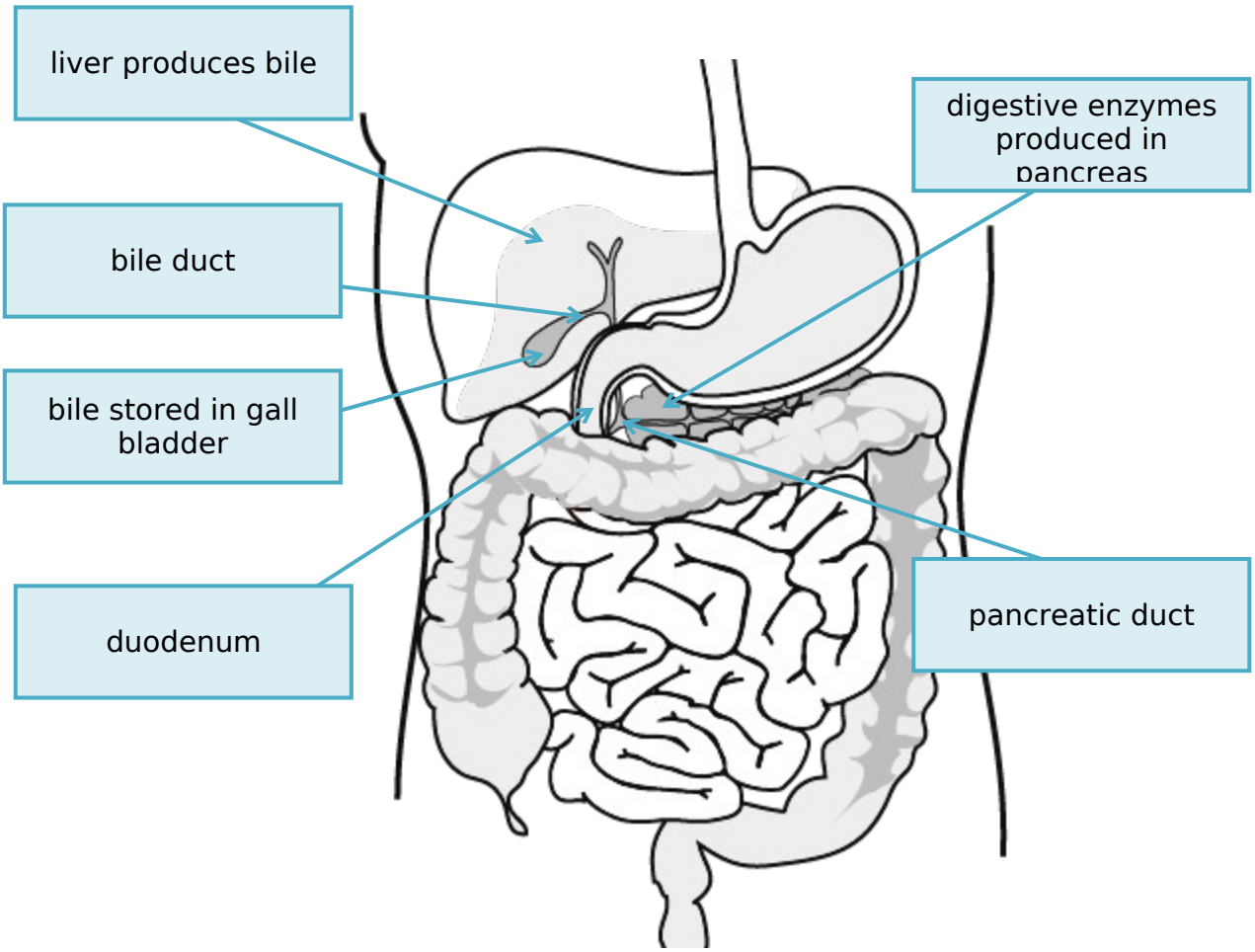
1. Suggest the name of the enzyme used in this experiment.
2. What are the products of the reaction?
3. What do you think will happen to the pH in each experiment?
4. Look at the table below. Compare and explain the results of the two experiments.

Time in minutes	pH	
	Experiment 1: with bile	Experiment 2: without bile
0	9.0	9.0
2	8.9	9.0
4	8.6	9.0
6	8.0	8.9
8	7.8	8.7
10	7.6	8.3

Teaching notes

Before doing Task 3 cut the sheet before question 4 so that this can be given to the students after they have answered the first 3 questions.

Suggested answers



Task 2

What role does bile have in digestion?

- It neutralises food, coming from the stomach, which is acidic.
- It provides an alkaline environment – which is the optimum pH for the enzymes which work in the small intestine
- It emulsifies fat – increasing the surface area of the fat exposed to lipase enzymes.

Task 3

1. Lipase
2. Fatty acids and glycerol
3. pH will go down as acids are produced by breakdown of fats. The pH will decrease at a faster rate in the first experiment.